



Faculty of Technology
Bachelor's Technology Degree Thesis

AGS

AQUAGUARD SYSTEM

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Abstract

The AquaGuard System (AGS) represents an advanced water tank management solution that utilizes cutting-edge IoT technology to automate and improve water control functions. The system combines real-time monitoring with automatic pump control and leak detection while integrating emergency response mechanisms alongside weather forecasting and security features to optimize water usage and prevent water loss while maintaining safety standards. AGS employs ultrasonic sensors for precise level measurement alongside turbidity sensors for water quality assessment and flow sensors for leak detection to produce essential data-driven insights. A connected mobile app enables users to perform remote water level monitoring and management while receiving immediate alerts and optimizing usage. AGS seeks to transform water management through its economical and dependable smart solution which enhances sustainability while minimizing operational waste across residential and commercial settings.

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List of abbreviations

Ph: POTENTIAL OF HYDROGEN
 NTU: NEPHELOMETIC TURBIDITY UNIT
 PWM: PUSLE WIDTH MODULATION
 MHz : MEGAHERTZ
 AC: ALTERNATING CURRENT
 DC: DIRECT CURRENT
 V: VOLTAGE
 I/P: INPUT
 mA: MILLIAMPERE
 KB: KILOBYTE
 mm: MILLIMETER
 g: GRAM
 TCP/IP: TRANSMISSION CONTROL PROTOCOL/ INTERNET PROTOCOL
 I2C: INTER-INTEGRATED CIRCUIT
 SPI: SERIAL PERIPHERAL INTERFACE
 UART: UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER
 MB: MEGABITE
 Ω : OHM'S
 m Ω : MILLIOHM
 NO: NORMALLY OPEN
 NC: NORMMALLY CLOSE
 KHz: KILO HERTZ
 V_{OUT} :: VOLTAGE OUTPUT
 TX: TRANSMIT
 RX: RECIVE
 °C: DEGREES CELSIUS
 ADC: ANALOG TO DIGITAL CONVERTER

L/MIN: LITER PER MINUTE

MPa: MEGAPASCAL

V_{IN} : VOLTAGE INPUT

PCB: PRINTED CIRCUIT BOARD

Chapter 1: Introduction

The Aquaguard System is right-oriented water management and purification system effectively monitoring and controlling the use of water. Various sensors are brought together in this system to operationally check conditions of water level, leakage detection, and water quality. Being connected to the internet, users can remotely track and manage water consumption. This system, when combined with pumps, can function in controlling them automatically so that the water supply will never be interrupted yet water wastage is kept to the minimum. It features instant alerts and automation that not only saves on manual efforts but help in saving water. The article shall give an insight on the working of the Aquaguard system, its features, and the benefits offered to modern water management... (Agarwal, January 2011)

Figure 1 Aquaguard system

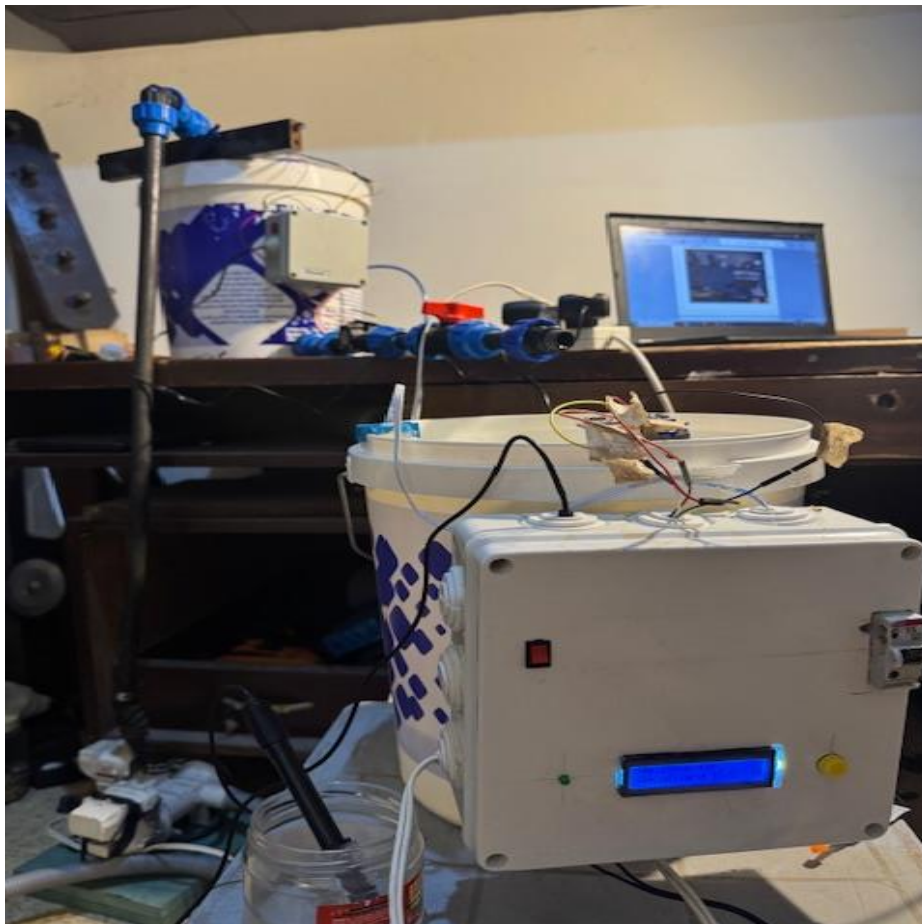


Figure 2

The Objective of Our Work

For our university's final project, we designed the AquaGuard: Intelligent Water Tank Management system. This work aims to produce an automated and intelligent water management system instead of some manual water management systems. First, the sensors measure water quality, level, and flow, enabling them to collect real-time data. The system then interprets the data to detect any abnormality that may be considered a water leakage or unauthorized use. Finally, it controls the water pumping operation, displays vital information on a mobile app, and alerts the user during emergencies or disturbances.

This study starts with the analysis of conventional water tank management systems and identifying their drawbacks with respect to efficiency work, safety, and automation. Then, tests on AquaGuard would be performed by engineering and end-user stakeholders to ensure that the system operates most optimally and satisfies all weed requirements. Thirdly, the evaluation of sensor performance shall be carried out for its capability to view the water level, turbidities, and leaks with accuracy. Fourthly, the feedback from these tests will be recorded and compared to those of existing solutions to evaluate the advantage of AquaGuard in terms of efficiency, safety, and ease of use.



Figure 3

Project Realization

This project will develop an intelligent water management system for water tanks operating in the residential and commercial sectors. AquaGuard monitors water quality, detects leaks, and waters plants efficiently. This system will be improved and rendered into better versions according to time, thus ensuring its continuous validity and scalability in different applications, following new technological advancements.



Figure 4

Chapter 2: Project's History

The demand for intelligent water management solutions led to the birth of AquaGuard. Manual testing and inspection were the basis of early water monitoring systems. Basic sensor -based systems that offered no real time data were introduced to automated water quality monitoring in early 2010s. But these early solutions lacked automation and remote connectivity.

Current water monitoring systems evolved with IoT and cloud computing, wireless connectivity, automated control and real-time data learning. By integrating multiple sensors like pH, turbidity, ultrasonic and water leakage sensors with ESP8266 and Arduino microcontrollers for smooth operation, AquaGuard builds upon these developments. Users can remotely monitor and control water levels with its integration with Blynk Cloud which ensures water conservation and minimization of waste.

Before AquaGuard, managing water required constant manual intervention which posed risk of harm, waste and overflow. This was completely changed with the creation of AquaGuard which gave an intelligent, automated and real time monitoring system. Today, these systems are needed for communities, businesses and homeowners to manage and protect water. (Siraparapu, December 2024)

Project's Plan

AquaGuard was born to solve big problems like water waste, manual monitoring and unacceptable resource usage due to the growing need for water management. Traditional water management systems rely on human intervention to monitor and control water levels which is a hard and expensive process.

By automating this process AquaGuard gives clear real time monitoring and eliminates manual labor. Reduces water loss and high electricity usage, this automation not only increases efficiency but also saves cost for businesses, municipalities and households. AquaGuard is a stable and affordable solution for current water management since it's connected to IoT technology, remote monitoring and control.



Figure 5

Project's Sampling:

The AquaGuard system is a prototype representation of a more advanced water management solution. Due to financial constraints, the project has not yet been implemented on a large scale. Instead, a smaller-scale version was developed to test the core functionalities and evaluate its effectiveness.

Promising results from early testing confirmed that the system can effectively detect leaks, automate motor control, and monitor water levels. However, the prototype was built with low-cost parts. The accuracy, stability, and general performance of the system would all be greatly improved by using better sensors and hardware, which would further increase its dependability for broad usage. (G. Caicedo-Escorcia, October 23, 2024)



Figure 6

Analyzing the Results of the AquaGuard System

Tests of the AquaGuard system revealed a tiny error margin in sensor readings of around 5% to 10%. Environmental factors such as polluted water, power interruptions, and nearby interference were the primary cause of these errors. In spite of this, the system set the motor automatically, detected leaks, and controlled water levels.

Feedback from users such as homeowners and facility staff showed positive. They found the system helpful and easy to use for managing water effectively. Most of the errors happened under test conditions, and the system is expected to perform even better in real-world situations. Overall, the results show that AquaGuard is a reliable and smart solution for water monitoring and control. (Jignesh Vidani, November 2023)

Data collection and analysis

Condition	Sensor Used	Sensor Accuracy	Motor/Relay State	Water Flow Control Efficiency	Blynk Alert Triggered	Alert Message Example
Roof Tank Water Level Low	Ultrasonic Sensor (Roof Tank)	95%	ON (Motor Activated)	100%	Yes	Roof Tank Low – Motor Started
Roof Tank Water Level High	Ultrasonic Sensor (Roof Tank)	98%	OFF (Motor Deactivated)	100%	No	—
Ground Tank Water Level Low	Ultrasonic Sensor (Ground Tank)	94%	OFF	100%	Yes	Ground Tank Empty – Check Water Source
Ground Tank Water Level OK	Ultrasonic Sensor (Ground Tank)	94%	Controlled Based on Logic	100%	No	—
Water Leakage Detected	Water Flow Sensor	92%	OFF	95%	Yes	Leak Detected – Turn Off Motor Immediately
Clean Water (Turbidity Normal)	Turbidity Sensor	93%	Controlled Based on Logic	100%	No	—
Dirty Water (High Turbidity)	Turbidity Sensor	90%	ON or OFF Based on Logic	98%	Yes	Water Dirty – Turbidity > 1100 NTU
pH Below Safe Range (pH < 6.5)	pH Sensor	89%	ON or OFF Based on Setup	95%	Yes	pH Low – Water is Acidic (pH < 6.5)
pH Above Safe Range (pH > 7.9)	pH Sensor	89%	ON or OFF Based on Setup	95%	Yes	pH High – Water is Alkaline (pH > 7.9)
pH Normal Range (6.5 – 7.9)	pH Sensor	89%	Controlled Based on Logic	100%	No	—
Low Power Supply	Voltage/Power Sensor (optional)	60% (if installed)	OFF	60%	Yes	Low Power – Switching to Safe Mode
System Operating Normally	All Sensors	>90% Avg	Controlled Based on Logic	100%	No	—
Blynk Disconnected	Wi-Fi / Blynk Module	—	OFF (Fail-safe Mode)	0%	Yes (Optional local alert)	Connection Lost – Check Wi-Fi or Cloud Connectivity

Table 1

Chapter 3: Design Process

The aim and objectives:

The purpose of this project is to offer a more efficient and precise method for monitoring and managing water reservoirs, quality, and safety. The primary aim is to conserve water, enhance safety, and reduce cost by increasing the efficiency of monitoring through real-time automation.

The AquaGuard system includes sensors and software that can measure levels of water, turbidity, leakage, and motor power consumption. From this data, Through this information AquaGuard manages to adjust water usage smartly while avoiding waste and ensuring steady water supply by regulating the pump and notifying users about issues.

Depending on the application, the AquaGuard system's accurate goals and objectives may change. For instance, automating water tank refilling and leak detection may be the aim in homes, whereas water usage optimization and production process quality control may be the main objectives in industrial settings.

General Process Flow Chart:

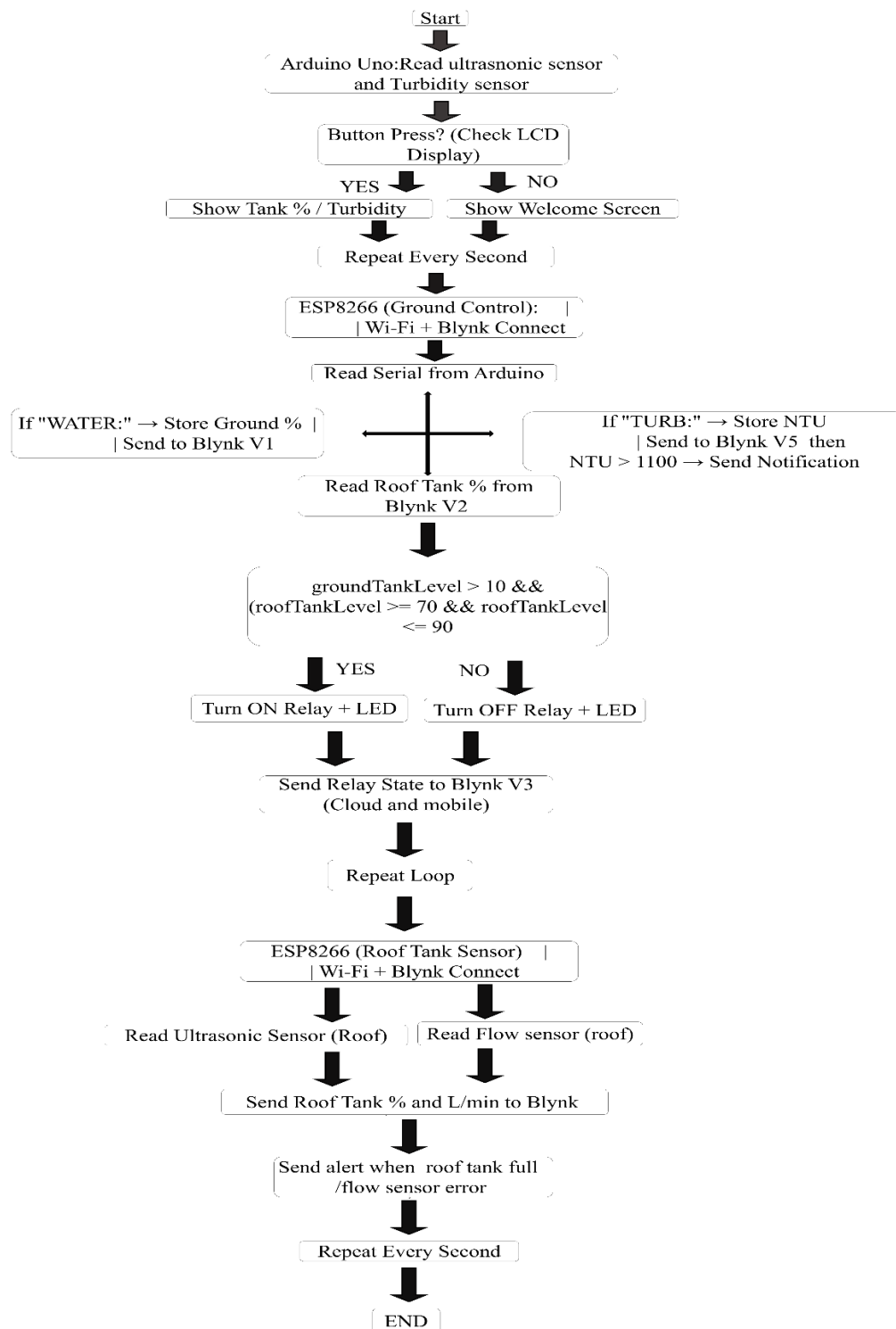


Figure 7: Flowchart aquaguard

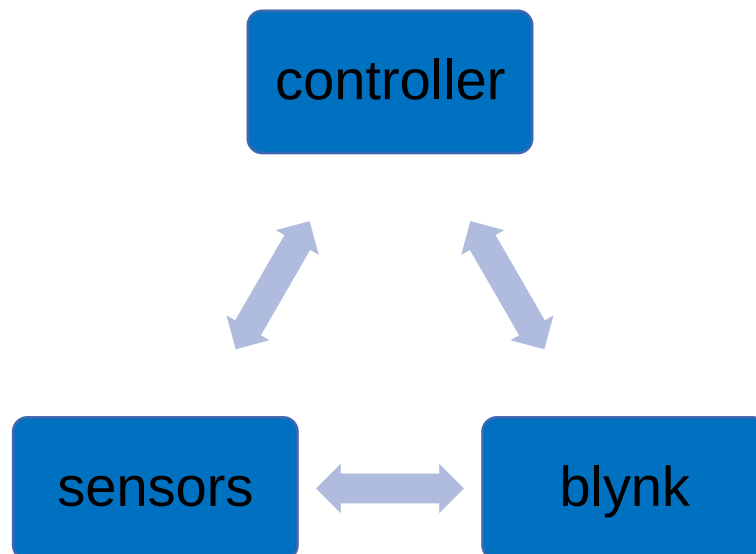
The pass:

The AquaGuard system's integrated sensors are intended to continuously check water quality and levels. The ultrasonic sensor measures the water level, while the turbidity sensor measures the water's clarity. upon the readings, the system automatically runs the motor to fill or empty the tank and alerts the user via the Blynk app. Also, if the water flow sensor detects any leaks, the system informs the user so that quick action can be done to stop water loss rapidly.

Chapter 4: Work Cycle

Study:

The AquaGuard system's ability to monitor water levels and quality depends on a steady power source as well as the correct components including sensors and preset rules. Results could be inaccurate if the system has no ability to read these inputs accurately. The system also needs a clear space free from outside interference, such as obstructions that could interfere with ultrasonic readings or problems interacting with Blynk via Wi-Fi. Alerts for turbidity, undetected leaks, or unclear water levels will only be sent by AquaGuard if there is a real trigger such undetected interference. (Ezz El-Din Hemdan, 24 February 2023)



Chapter 5: Project description

The project divided into five sections:

1. Software
2. Electronic control
3. Mechanical
4. Webpage and mobile app
5. PCB Design

Software:

In this design, the microcontrollers used are the Arduino Uno and the ESP8266. Both were programmed using the Arduino IDE with C++ language. It connects to Blynk Cloud.

Electronic components:

Arduino Uno R3

Arduino UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started. You can tinker with your UNO without worrying too much about doing something wrong, worst-case scenario you can replace the chip for a few dollars and start over again (Arduino, n.d.)



Figure 8:Arduino Uno R3

Specification:

- It is an ATmega328P based Microcontroller
- The Operating Voltage of the Arduino is 5V
- The recommended input voltage ranges from 7V to 12V
- The I/p voltage (limit) is 6V to 20V
- Digital input and output pins-14
- Digital input & output pins (PWM)-6
- Analog I/p pins are 6
- DC Current for each I/O Pin is 20 mA
- DC Current used for 3.3V Pin is 50 mA
- Flash Memory -32 KB, and 0.5 KB memory is used by the boot loader
- SRAM is 2 KB
- EEPROM is 1 KB
- The speed of the CLK is 16 MHz
- In Built LED
- Length and width of the Arduino are 68.6 mm X 53.4 mm
- The weight of the Arduino board is 25 g

ESP8266

The cheap Wi-Fi microchip with full TCP/IP stack and microcontroller functionality is the ESP8266. It supports several communication protocols, including I2C, SPI, UART, and PWM, and it has 17 GPIO pins. It can be powered by external sources or via USB. It is perfect for projects that need internet connection given that it is made for the use in Internet of Things applications and makes Wi-Fi connectivity simple. The Arduino IDE may be used to program the ESP8266, making it easier to integrate with other sensors, actuators, and cloud platforms. (Espressif, n.d.)



Figure 9:Esp8266 Nodemcu

Specification:

- Microcontroller: Tensilica Xtensa 32-bit LX106
- Operating Voltage: 3.3V (Logic level)
- Input Voltage: 3.3V (recommended), 5V (max)
- Digital I/O Pins: 17 (varies with module variant)
- PWM Pins: 9 (varies with module variant)
- Analog Input Pins: 1 (10-bit ADC, A0)
- DC Current per I/O Pin: 12 mA (recommended)
- Flash Memory: 4MB (varies with module variant)
- SRAM: 64 KB
- EEPROM: 512 KB
- Clock Speed: 80 MHz (default, can be overclocked to 160 MHz)
- Built-in LED: Yes (on GPIO2 or other, varies by module)
- Wireless Standard: IEEE 802.11 b/g/n Wi-Fi
- Wi-Fi Features: Wi-Fi Direct, TCP/IP stack
- Length: Varies (typically 24.5mm x 16mm for ESP-01)
- Width: Varies (typically 16mm for ESP-01)
- Weight: Approx. 3-4 g (depends on the module version)

Push Button

A push button is a simple, manually operated switch used to make or break an electrical connection. It is typically used in control circuits or embedded systems to provide user input). The most common form is a momentary push button, which only stays activated while being pressed. (Pola Risma, February 2020)

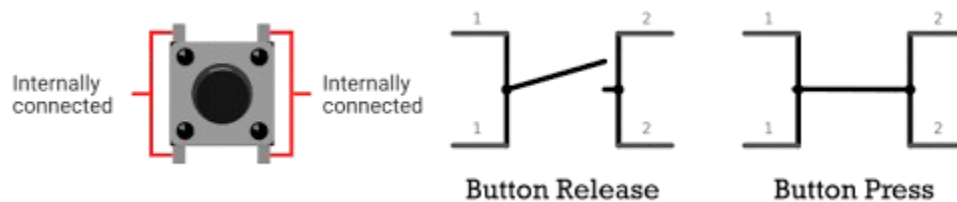


Figure 10: PushButton

Specifications (Typical for 4-pin Push Button):

- Type: Momentary switch (normally open)
- Number of Pins: 2 or 4 (internally connected in pairs)
- Operating Voltage: 3.3V to 5V (logic level compatible)
- Current Rating: 50 mA to 100 mA (low current)
- Contact Resistance: $< 100 \text{ m}\Omega$
- Mechanical Life: $\sim 100,000$ presses or more
- Mounting: Through-hole or surface mount (THT/SMD)
- Dimensions: Vary, standard size is $\sim 6\text{mm} \times 6\text{mm}$ for small tact switches

How it works:

- Internally, the push button consists of two metal contacts separated by a spring mechanism.
- When the button is not pressed, the circuit is open and no current flows.
- When the button is pressed, the contacts are pushed together, closing the circuit and allowing current to flow.

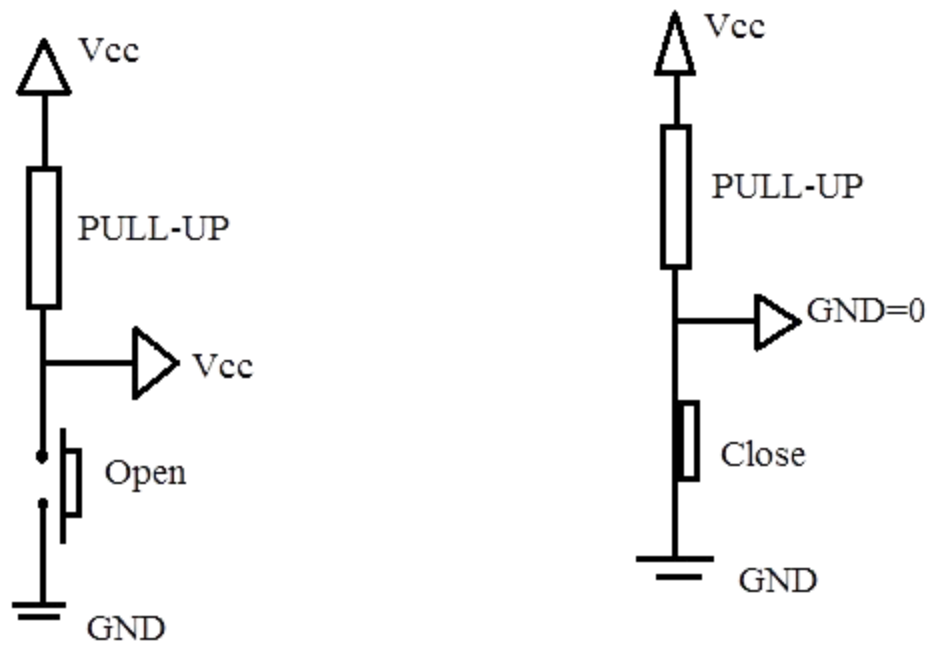


Figure 11 Circuit of push button

LED:

LCD1602 IIC/I2C blue black-light

This is I2C interface 16x2 LCD display module, a high-quality 2-line 16-character LCD module with on-board contrast control adjustment, backlight and I2C communication interface. For Arduino beginners, no more cumbersome and complex LCD driver circuit connection. The real significance advantages of this I2C Serial LCD module will simplify the circuit connection, save some I/O pins on Arduino board, simplified firmware development with widely available Arduino library. (John Doe, 2023)



Figure 12: LCD1602 IIC/I2C blue black-light

Specification and features:

- Arduino IIC/I2C interface was developed to reduce the IO port usage on Arduino board
- I2C adapter allows flexibility in connections
- I2C reduces the overall wirings.
- 16 characters wide, 2 rows
- White text on the blue background
- Single LED backlight included can be dimmed easily with a resistor or PWM.
- Interface: I2C
- Interface Address: 0x27
- Character Color: White
- Backlight: Blue
- Supply voltage: 5V
- Overall dimensions: 3.15 x 1.42 x 0.51 in (80 x 36 x 13 mm)
- Character size: 2.95 mm wide x 4.35 mm high

How it works

It initially shows the signal of each color, and when the color sensor coincides with the desired color, the panel records an increase in the number on the color that the robot transfers, and when 12 pieces are reached, the start of counting appears on the screen again and a number is added to the group for the desired color to the group for the desired color (R, G, B)

LED:

Why we use led in Aquaguard system :

- When the motor is ON, the LED turns ON = indicating that the water pump is actively running.
- When the motor is OFF, the LED is OFF = indicating that the pump is not operating.

how it works :

applied to Ohm's Law:
$$I = \frac{V_{ESP} - V_{LED}}{R}$$

where:

Output voltage of ESP8266 GPIO pin (3.3V)

LED = Forward voltage of the LED (e.g., 2V for red, ~3V for white/blue)

R = Resistor in series with the LED (e.g., 220Ω)

I = LED current in amperes (A) or milliamperes (mA)

Relay Module 1 channel

An electronic switching device called a 1-Channel Relay Module is used with low-voltage microcontrollers such as Arduino or ESP8266 to regulate high-voltage AC or DC circuits. It functions as an electrical switch that allows you to safely turn on and off devices including lights, fans, pumps, and motors.

The module shields the microcontroller from high-voltage spikes by using an optocoupler for electrical isolation. The relay changes its state, allowing or disabling power to the linked load, in response to a signal from the microcontroller to the control pin.

When protect switching and electrical isolation are needed for industrial control systems, IoT projects, and home automation, this module is perfect. (Hameed, 2020)



Figure 13: Relay module 1 channel

Specification:

- Operating Voltage: 5V
- Control Voltage: 3.3V or 5V
- Relay Type: SPDT or DPDT
- Maximum Switching Voltage: AC:250V AV , DC:30V DC
- Maximum Switching Current: AC:10A , DC:10A
- Trigger Type: Low-level trigger

- Control Pin Current: 10-15 mA
- Output Contact: Normally Open and Normally Closed pins
- Relay Coil Resistance: 70-100Ω
- Inductive Load Capacity: Suitable for driving motors, lights, fans, or any high-voltage device
- Isolation: Optocoupler
- Built-in Protection: Diode for flyback protection
- Mounting: 4 pins for control (VCC, GND, IN, and COM) and 3 pins for the relay's NO, NC, and Common contacts

How it works:

1. Triggering the Relay: When a voltage is applied to the "IN" pin, the relay gets energized.
2. Switching Mechanism: Once energized, the internal switch connects the NO pin to the Common pin, closing the circuit and allowing current to flow to the connected load.
3. Deactivation: When the control voltage is removed, the relay de-energizes, and the internal switch connects the NC pin to the Common pin, breaking the circuit and stopping the current flow.

Physics Behind Relay Operation:

1. Current in Coil → Magnetic Field (Ampère's Law)

When ESP8266 sets the control pin LOW, a current flow through the relay's coil:

$$I = \frac{v}{R} = \frac{5}{70} = 0.071A$$

The coil generates a magnetic field:

$$B = \mu_0 \times \frac{N \times I}{L} = 1.785mT$$

Where: $\mu_0 = 4\pi \times 10^{-7} T.m/A$

N=400 turns (typical)

L= 2cm = 0.02m

2. Electromagnetic Force in Relay – Applied Calculation:

Formula:
$$F = \frac{N \cdot I \cdot B \cdot l}{L}$$

Where:

- F = Magnetic force (Newtons)
- N = Number of turns in the coil
- I = Current through the coil (Amperes)
- B = Magnetic field strength (Tesla)
- l = Length of conductor inside the magnetic field (meters)
- L = Magnetic path length (meters)

Parameter	Value	Notes
N	400 turns	Estimated for relay coil
I	0.071 A	From 5/70
B	0.0018 T (1.8 mT)	From Ampère's Law
l	0.01 m (1 cm)	Length of coil exposed to magnetic field
L	0.02 m (2 cm)	Magnetic path length (core loop length)

Substitute into formula:

$$F = \frac{400 \cdot 0.071 \cdot 0.0018 \cdot 0.01}{0.02}$$

$$F = 0.02556\text{N}$$

Conclusion: That's 25 millinewtons, which is more than enough to pull the lightweight iron armature and close the contacts

Truth Table for Relay Control:

Ground Tank Level	Roof Tank Level	Relay State (Motor)	LED
≤ 10	Any value	OFF	OFF
> 10	< 70	OFF	OFF
> 10	> 90	OFF	OFF
> 10	70 to 90 (inclusive)	ON	ON

Table 2

Logic level shifter:

A Logic Level Shifter is an electronic circuit or module used to safely convert signals from one voltage level to another. It allows devices operating at different logic levels (for example, a 5V Arduino and a 3.3V ESP8266) to communicate without damaging the lower-voltage components.

Level shifters are commonly used in mixed-voltage systems to ensure reliable data exchange between microcontrollers, sensors, modules, or peripherals operating at different voltage standards. (Johanneck, (2021, November 4))

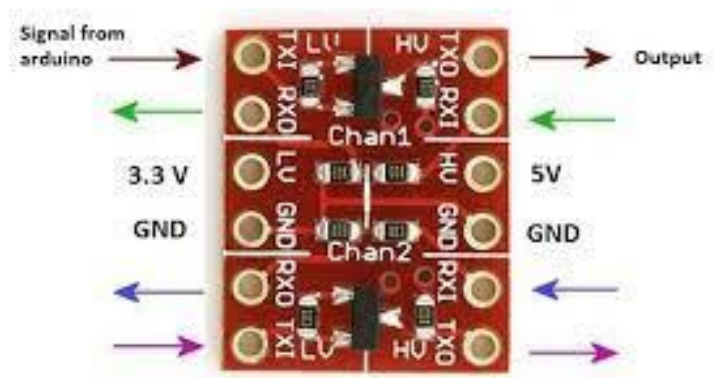


Figure 14: Logic Level Shifter

Schematic diagram

10k3.3V 5V TTL Bi-directional Logic Level Converter

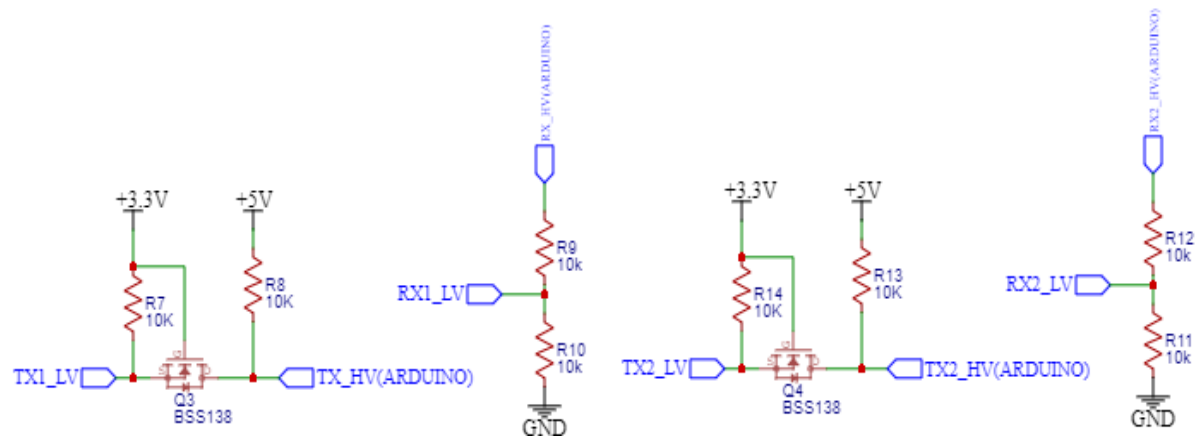


Figure 15 circuit Bi-directional Logic level converter

Specification:

- Input Voltage High Side: 3.3V to 5V
- Input Voltage Low Side: 1.8V to 3.3V
- Maximum Current per Channel: 50 mA (varies with component type)
- Supported Data Rates: Up to 100 kHz (for I2C) / Several MHz (for SPI)
- Number of Channels: 2, 4, 8, or more (varies by module)
- Type: Bi-directional or Uni-directional
- Isolation: No (direct connection with voltage translation)

How it works:

A logic level shifter typically uses MOSFET transistors or dedicated ICs to detect the voltage level on one side and adjust the output to match the required logic level on the other side.

- When a high logic signal (for example, 5V) is sent from the high-voltage side, the level shifter converts it to a safe 3.3V signal for the low-voltage device.
- When a low logic signal (0V) is sent, it remains at ground level, compatible for both sides.

This allows reliable bi-directional or uni-directional communication between devices with different logic voltages.

Voltage Divider:

$$\text{Formula: } V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$$

Where:

- $V_{in}=5V$ (Arduino TX)
- $R_1=10k\Omega$ (top resistor, connected to Arduino TX)
- $R_2=20k\Omega$ (bottom resistor, connected to ESP8266 RX and GND)

$$V_{out} = V_{out} = 5 \times \frac{20}{10+20} = 5 \times \frac{2}{3} \approx 3.33v$$

Transistor:

In the AquaGuard system, the transistor (2N2222) acts as an electronic switch that allows the low-power ESP8266 microcontroller to control higher-current components like the **relay** module. This relay, in turn, operates the water pump based on signals from the system (such as tank water levels).

By using the transistor, the ESP8266 can safely and effectively activate the pump without being overloaded.

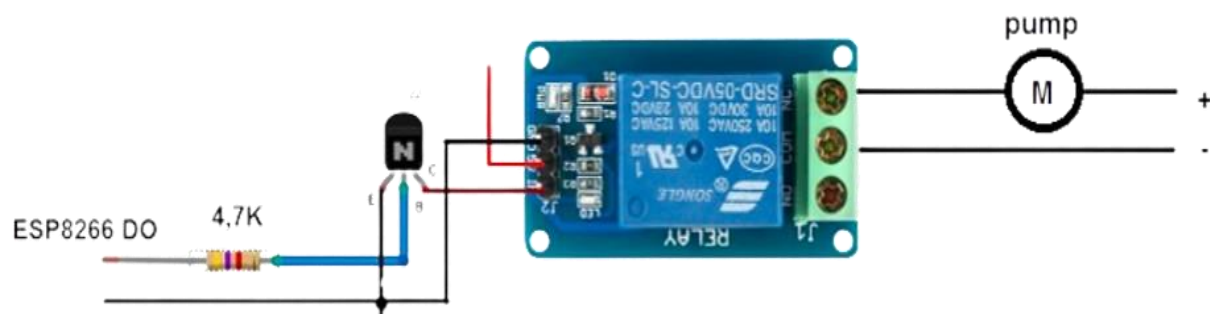


Figure 16 Transistor

Key Reasons:

1. Current Limitation of ESP8266:

- The GPIO pins on the ESP8266 can only supply a small amount of current (typically ~12 mA max, safely).
- A relay coil often requires 70–100 mA, which exceeds the ESP8266's capability.

2. Voltage Compatibility:

- Many relays are designed for 5V logic, but the ESP8266 operates at 3.3V on its GPIO pins.
- The transistor (e.g., 2N2222) acts as a switch that uses the 3.3V signal to control a 5V current.

3. Protection:

- Switching a relay can cause voltage spikes (back EMF) that may damage the ESP8266.
- The transistor acts as a buffer, and a diode (usually across the relay coil) is used to protect against these spikes.

Ultrasonic sensor:

An electronic device called an ultrasonic sensor uses ultrasonic sound waves to determine how far away an item is. It measures the time it takes for an echo to return after reflecting off a nearby object after producing a sound pulse at a frequency higher than human hearing, typically 40 kHz.

Automation, robotics, level measurement (such as in water tanks), and object detection systems all make considerable use of these sensors. (Saleem Latteef Mohammed, November 28, 2019)

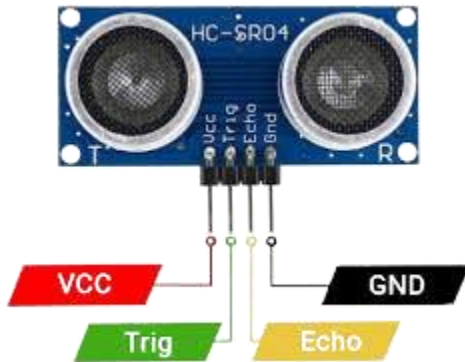


Figure 17: Ultrasonic Sensor

How it works:

The ultrasonic sensor operates using the time-of-flight principle:

1. The transmitter (Trigger pin) sends out an ultrasonic sound pulse.
2. The sound pulse travels through the air until it hits the surface of the water.
3. The sound wave bounces back and the receiver (Echo pin) detects the returning echo.
4. The sensor measures the time taken for the pulse to travel to the water and back.
5. Using the known speed of sound (approximately 343 m/s), the system calculates the distance to the water surface.

Formula:

$$\text{Water level}(\%) = \left(\frac{\text{Maximum Tank Depth} - \text{Measured Distance}}{\text{Maximum Tank Depth}} \right) \times 100\%$$

$$\text{Water level}(\%) = \left(\frac{30-10}{30} \right) \times 100 = 66.7\%$$

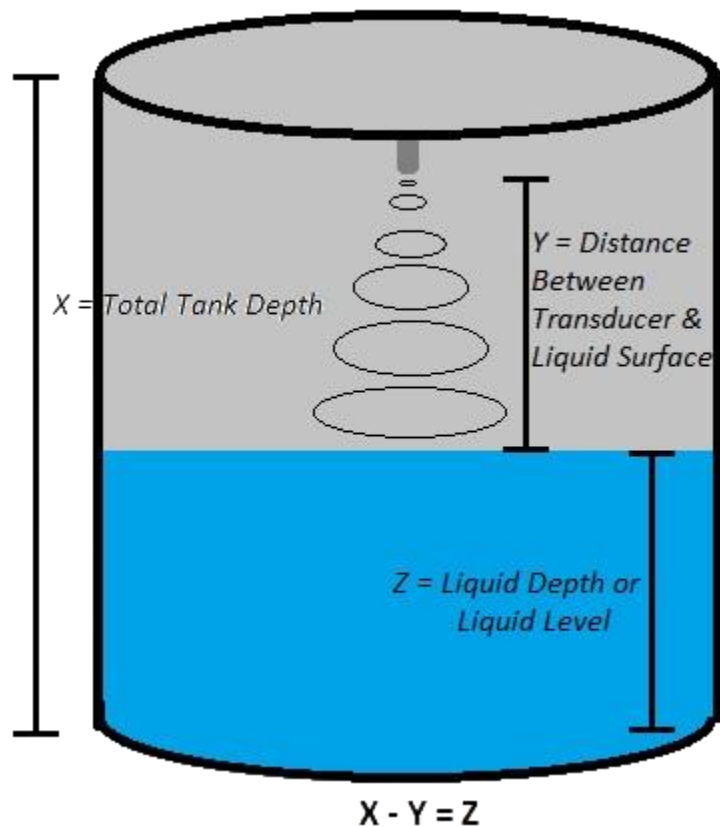


Figure 18:how it works

Typical Specification:

- Operating Voltage: 5V DC
- Operating Current: 15 mA
- Frequency: 40 kHz
- Measuring Range: 2 cm to 400 cm (0.02 m – 4 m)
- Measuring Angle: $\pm 15^\circ$
- Resolution: ~ 1 mm
- Trigger Input Signal: 10 μ s TTL pulse
- Echo Output Signal: Pulse width proportional to distance
- Operating Temperature: -15°C to $+70^\circ\text{C}$
- Dimensions: Approx. 45mm x 20mm x 15mm (for HC-SR04 module)

Turbidity sensor

A turbidity sensor measures the cloudiness or haziness of a liquid, which indicates the amount of suspended solids present in the water. In water systems, high turbidity usually means contamination or particles are floating in the water. The sensor helps detect whether the water is clean or dirty by measuring how much light passes through the water sample.

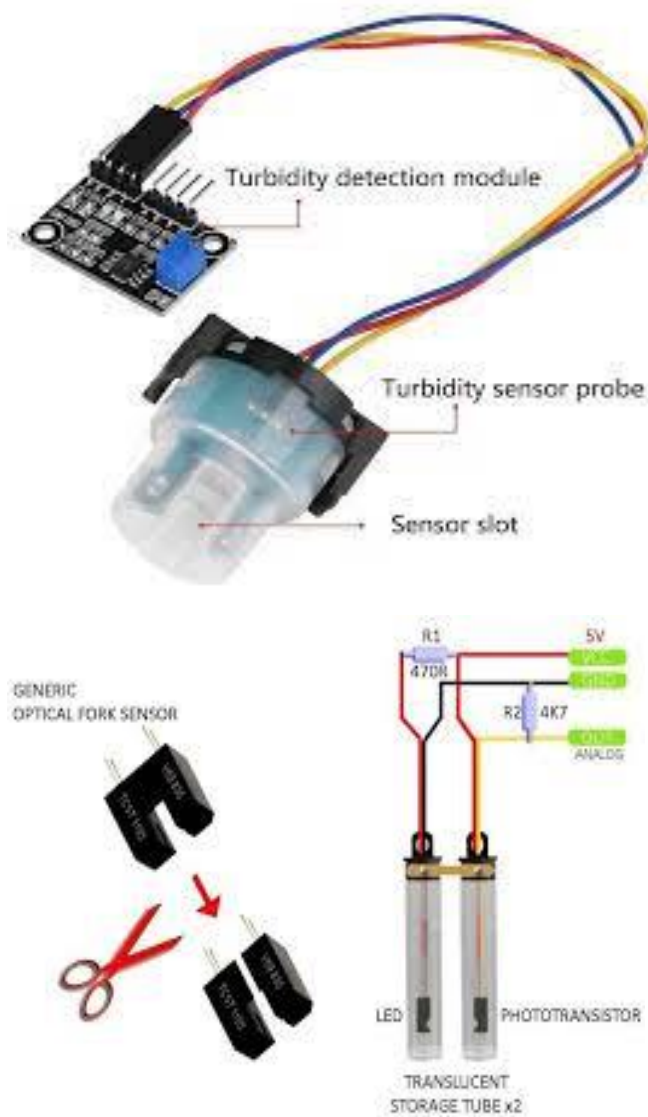


Figure 19: Turbidity sensor

Working principle:

The turbidity sensor operates based on light scattering:

1. The sensor has a light emitter (usually an infrared LED) and a light detector (phototransistor).
2. When the water is clear, most of the light passes straight through with little scattering.
3. When the water is cloudy or dirty, suspended particles scatter the light, reducing the amount reaching the detector.
4. The sensor outputs an analog voltage signal that varies according to the water's turbidity level — the dirtier the water, the higher the change in voltage.

Formula:

Beer–Lambert Law Recap:

$$I = I_0 \cdot e^{-k \cdot x}$$

Where:

- I = Final light intensity (after water)
- I_0 = Initial light intensity (before water)
- k = Turbidity coefficient (depends on particle concentration)
- x = Distance light travels through water (cm or mm)

$$I = 100 \cdot e^{-0.3 \cdot 1.5}$$

$$e^{-0.5} \approx 0.6376$$

$$I = 100 \cdot 0.6376 \approx 63.76$$

We conclude: Light intensity detected by the sensor is approximately **63.76%** of original.

Formula:

1. Raw Analog to Voltage Conversion: $voltage = \frac{turbidityraw \times 5}{1023}$
2. Voltage to NTU Mapping: $turbidity\ NTU = \text{map}(voltage \times 100 \times 500 \times 3000)$

Calculation methods:

Calculate for $NTU < 1100$

$NTU = \text{map}(voltage \times 100 \times 500 \times 3000)$ means:

$Turbidity\ NTU = 3000 - (x * 6)$

so: $x = voltage * 100 = 3000 - (x * 6) < 1100$

$$= x > \frac{(3000 - 1100)}{6}$$

$$= x > 316.67$$

That mean $= voltage * 100 > 316.67$

$= voltage > 3.167V$

Convert Voltage to Analog Value (ADC)

Now we convert that voltage to analog Read value:

Analog Value $= voltage \left(\frac{1023}{5} \right)$

Analog Value $= 3.167 \left(\frac{1023}{5} \right) \approx 648\ NTU$

Final Result of NTU :

Quantity	Value
Voltage	$> 3.167\ V$
analogRead	> 648

Table 3

Expression and Meaning:

Expression	Meaning
Turbidity Raw	The analog sensor reading from pin A0 (ranges from 0 to 1023).
5	The reference voltage of the Arduino (maximum possible voltage on analog pin).
1023	The maximum ADC value on Arduino (10-bit ADC = $2^{10} - 1 = 1023$).
$\frac{\text{turbidityraw} \times 5}{1023}$	Converts the raw ADC value to actual measured voltage in volts.
voltage×100	Converts the voltage to a scaled unit for easier mapping (for example, 4.56V becomes 456)
map (voltage × 100 × 500 × 3000)	Maps the scaled voltage from a range of 0–500 (equivalent to 0–5V) to a turbidity NTU value from 3000 NTU (clean) to 0 NTU (dirty).

Table 4

Specification and parameter

parameter	Specification
Detection Range	0 – 1000 NTU (typical for water)
Output Type	Analog Voltage Output
Power Supply Voltage	5V DC
Typical Clean	Water Output ~ 4.5V – 4.7V (Low turbidity, ~0 NTU)
Typical Dirty	Water Output ~ 2.5V – 3.0V (High turbidity, ~1000 NTU)
Operating Temperature	-30°C ~ 80°C
Response Time	~ 500ms
Stability	± 5%
Output Resolution	0.01V steps (ADC limited)
Wiring	VCC (Red) → 5V, GND (Black) → GND, AOUT (Blue) → A0

Table 5

Water flow sensor (YF S-201)

A small and dependable water flow sensor for determining the speed at which water moves through a pipe is the YF-S201. IoT-based water monitoring systems like your AquaGuard, automatic water dispensers, and smart irrigation all frequently use it.

A rotor (wheel), a hall-effect sensor, and a plastic body make up the sensor. The rotor rotates when water passes through the body. The hall sensor generates a digital pulse signal proportional to the flow rate after detecting the rotation speed. (Components101, 2021, May 19)

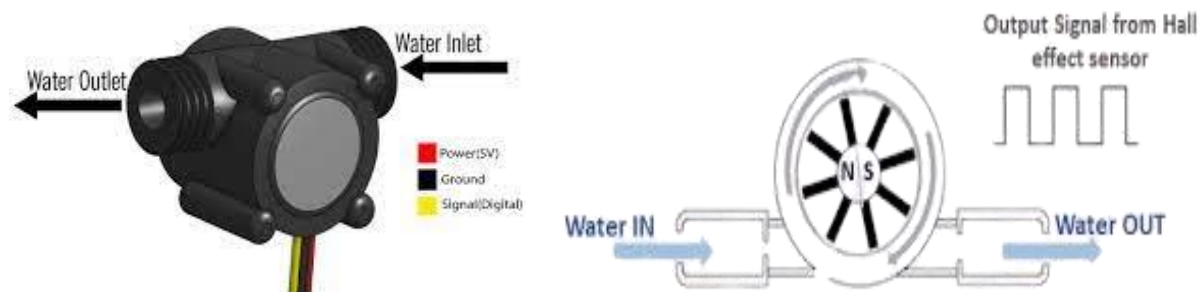


Figure 20:YF-S201

How it works:

1. Water Flow: When water enters the sensor, it passes through the chamber and spins the internal rotor.
2. Magnetic Detection: The rotor has a magnet attached to it. Each time the rotor completes a partial or full turn, the Hall Effect sensor detects the magnetic field change.
3. Pulse Output: For every spin, the sensor sends out a pulse signal to your microcontroller (Arduino). The number of pulses per second is directly related to the flow rate.

Physics Concepts Applied:

- Continuity Equation: $Q = A \cdot v$

Where:

- Q = Flow rate (m^3/s)
- A = Cross-sectional area of the pipe (m^2)
- v = Flow velocity (m/s)

Turbine Dynamics:

- The water exerts tangential force on turbine blades.
- This force causes rotation: $\tau = r \cdot F_{\text{fluid}}$

Where:

- τ = Torque
- r = Radius of the turbine
- F_{fluid} = Force from water flow

General Formula

$$\text{Flow Rate (L/min)} = \left(\frac{\text{Pulse Frequency(Hz)}}{\text{Calibration Factor}} \right) = \frac{75}{7.5} = 10 \text{ L/min}$$

Explanation formula:

Components	Meaning
Pulse Frequency	The number of pulses counted in 1 second.
7.5	The calibration factor (typical for YF-S201) — meaning: 7.5 pulses per second = 1 L/min.
Flow Rate	Calculated water flow rate in Liters per minute.

Table 6

Specification:

Parameter	Specification
Model	Yfs201
Operating Voltage	5V – 18V DC
Operating Current	≤ 15 mA (at 5V)
Flow Range	1 – 30 Liters per Minute (L/min)
Working Pressure	≤ 1.75 MPa
Output Signal	Digital Pulse (Hall Effect Sensor)
Connection	3-Pin: + (Red) / GND (Black) / Signal (Yellow)
Thread Size	1/2 inch (typical for standard water pipes)
Accuracy	±10% (under calibrated flow)
Material	Plastic (typically nylon) and Copper/Brass core
Temperature Range	-25°C to +80°C
Lifespan	≥ 300,000 Cycles

Table 7

PH sensor:

The acidity or alkalinity of a liquid solution can be determined using an electronic instrument called a pH meter sensor. Often wrapped into a single item called a combination pH electrode, it is composed of a pH electrode (probe) and a reference electrode. The meter displays the pH value after the sensor transforms

the electrical signal it obtains from the detection of hydrogen ion activity in the solution. (Khan)



Figure 21 ph sensor

Specification and parameter:

Parameter	Specification Range
Measurement Range	0 to 14 pH
Accuracy	± 0.01 to ± 0.1 pH
Resolution	0.01 pH
Response Time	< 1 to 30 seconds
Temperature Range	0°C to 80°C (some up to 100°C)
Electrode Type	Glass membrane (most common)
Reference Electrode	Ag/AgCl (Silver/Silver Chloride)
Output Signal	Analog (mV) or Digital (RS485, I2C, etc.)
Calibration	2-point or 3-point (pH 4, 7, 10 buffers)
Lifetime	1 to 3 years (depends on usage & maintenance)

Linear Calibration Equation:

the relationship between voltage and pH is linear, so we use the equation:

$$pH = m \times voltage + b$$

Where:

- m is the slope
- b is the intercept

Calculation: Let's call:

- $V_1=3.145$, $pH_1=4.01$
- $V_2=2.627$, $pH_2=6.86$

Slope m :

$$m = \frac{PH_2 - PH_1}{v_2 - v_1} = \frac{6.86 - 4.01}{2.627 - 3.145} = \frac{2.85}{-0.518} \approx -5.5019$$

Intercept b :

$$b = PH_1 - m \times V = 4.01 - (-5.5019 \times 3.145) = 4.01 + 17.301 \approx 21.311$$

Final Calibration Equation:

$$PH = -5.5019 \times Voltage + 21.311$$

Sample Source	Analog	Voltage (V)	pH	Comment
Distilled Water	575	2.808	~6.0	Slightly acidic (normal)
Buffer pH 4.01	644	3.145	4.01	Used for calibration
Buffer pH 6.86	538	2.627	6.86	Used for calibration
Ground Tank Water	503	2.456	7.80	Safe and alkaline

Table 8:

Graph of pH Sensor Calibration

The graph shows the linear relationship between voltage readings from the pH sensor and the known pH values of two buffer solutions (4.01 and 6.86). This calibration curve was used to determine the equation for converting voltage to pH in the AquaGuard system.

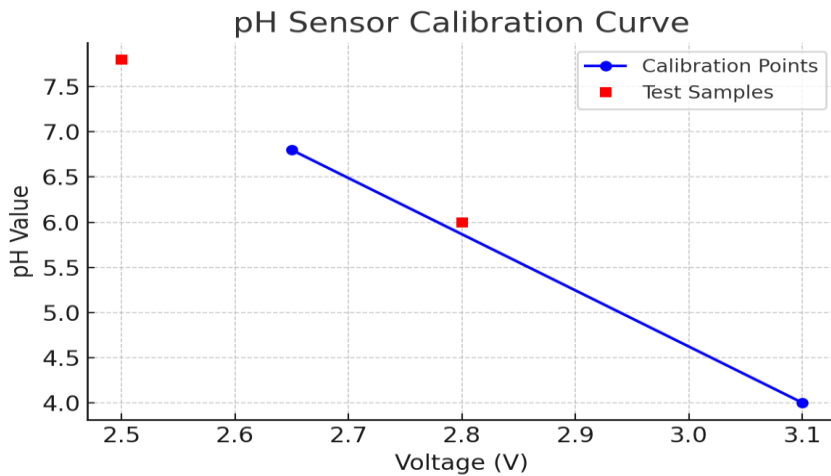


Figure 22 ph sensor calibration curve

LED:

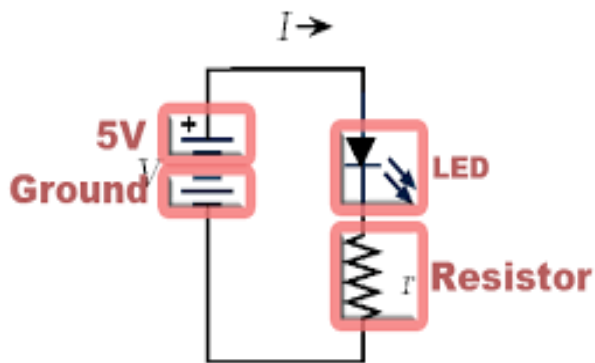


Figure 23LED

- When the motor is ON, the LED turns ON = indicating that the water pump is actively running.
- When the motor is OFF, the LED is OFF = indicating that the pump is not operating

LED Current-Limiting Resistor Calculation:

Applied to Ohm's Law
$$R = \frac{V_{SUPPLY} - V_{LED}}{I_{LED}}$$

Assumptions:

- Supply voltage $V_{SUPPLY} = 5V$ (from ESP8266 or relay module)
- LED forward voltage $V_{LED} = 2V$ (typical red LED)
- Desired LED current $I_{LED} = 20mA = 0.02A$

Calculation: $R = \frac{5V - 2V}{0.02A} = 150\Omega$

Power Dissipated by the Resistor: To make sure the resistor can handle the power,

calculate: $P = I^2 \times R = (0.02)^2 \times 150 = 0.06 W$

Mechanical Components

This section provides an explanation and introduction to the mechanical and external body components of the AquaGuard system. It covers the basic principles, the names of the components, their physical specifications, and the functional role of each part within the project.

Water pump Pedrollo ½ Inch

A modest and effective water pump, the Pedrollo 1/2 inch water pump is intended for small-scale and residential water transfer applications. It is in charge of transporting water between the AquaGuard system's tanks, ensuring dependable flow and pressure regulation. This pump has become known for its dependability, energy economy, and low maintenance requirements for clean water handling. Because of its 1/2-inch inlet and outlet connectors, it can be easily integrated into the system's plumbing network and is compatible with conventional pipe fittings. (Pedrollo., n.d)



Figure 24: Water pump

Water pipe fitting and hoses

The AquaGuard system's fluid transport network is centered on water pipe fittings and hoses. These parts allow safe and waterproof connections between the Pedrollo pump, sensors, and water tanks. The system provides easy maintenance and smooth water flow by using standard-sized hoses and connectors that fit the pump's 1/2-inch input and exit. While the hoses provide secure and reliable water transmission even in the presence of variable pressure conditions, the fittings offer flexibility in installation and routing. To preserve system efficacy and stop water leaks, their proper placement and selection are essential. (Depot., n.d.)



Figure 25 Water pipe fitting and hoses

Junction box waterproof

In order to keep the AquaGuard system's electronic components from adverse weather conditions, waterproof enclosures are essential. In order to ensure safe and steady operation, these enclosures are made to protect delicate modules—like microcontrollers, relays, and power supplies—from dust, moisture, and direct water contact. Using waterproof enclosures improves safety and extends the system's lifespan, particularly in outdoor or wet environments where access to water is expected. (Polycase., n.d.)

Two size of junction box:

Size:100x100x70



Size:225x200x80



Figure two junction box

Ultrasonic Bracket/housing

The housing and brackets for the ultrasonic sensors are made to solidly install them in the proper location for exact water level readings. These housings assure the measuring direction is correctly aligned with the water's surface, offer mechanical stability, and cover the sensors from damage. Although the brackets are often composed of materials that are resistant to corrosion, they will withstand the moist and wet conditions found inside tanks while still providing accurate sensor readings. (Makerfabs., n.d.)

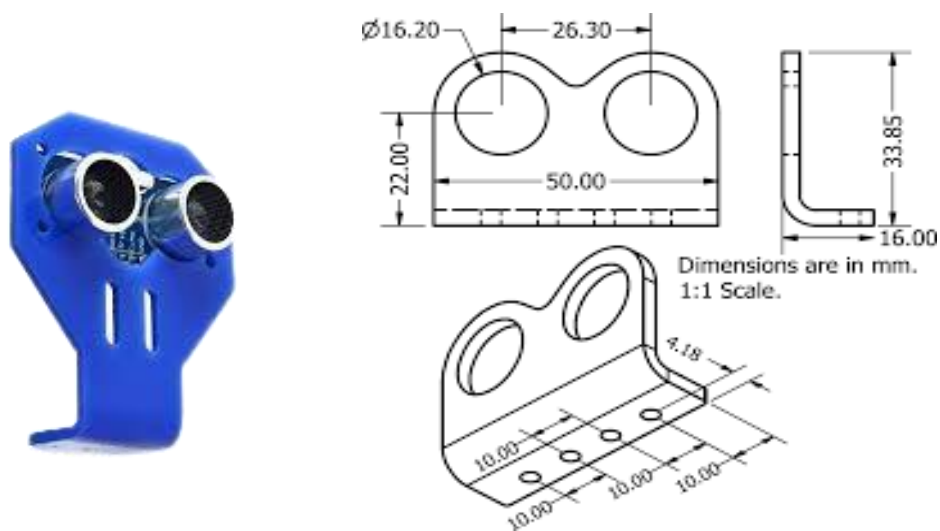


Figure 26 Ultrasonic Bracket/housing

Blynk Cloud and Mobile Application Integration

Overview:

The Blynk Cloud and Blynk Mobile Application are integrated to improve the AquaGuard system's remote monitoring and control abilities. With the help of a simple mobile interface, this function makes it possible to visualize data in real time and interact with the system without requiring direct physical contact. (Blynk., n.d.)

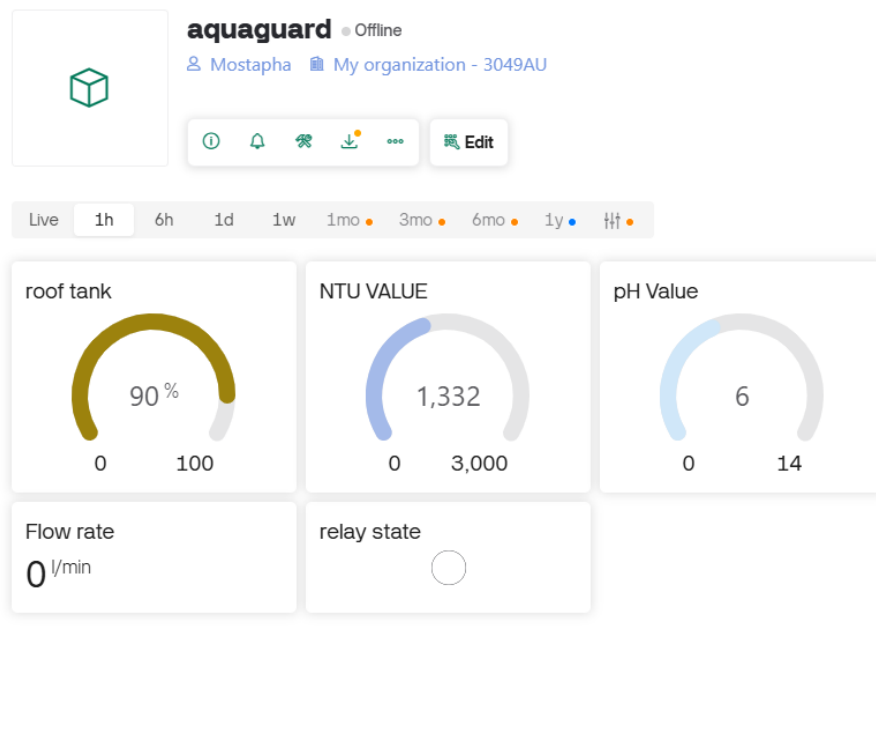


Figure 27 Blynk Cloud and Mobile Application Integration

Purpose:

The integration of Blynk Cloud and the Blynk Mobile App allows the AquaGuard system to:

- Send real-time sensor data to the cloud.
- Display water level and turbidity values on a mobile device.
- Enable users to monitor water conditions remotely from anywhere.
- Support future features like notifications and remote control actions (e.g., turning pumps on/off).

How it works:

1. Data Collection:

- The Arduino UNO reads values from two sensors:
- The ultrasonic sensor measures the water level.
- The turbidity sensor measures water quality.
- The processed sensor data is sent to the ESP8266 microcontroller via serial communication.

2. Wi-Fi and Cloud Communication:

- The ESP8266 (NodeMCU) connects to a Wi-Fi network and establishes a secure connection with Blynk Cloud using a unique authentication token.
- The ESP8266 transmits the received data to the Blynk Cloud using virtual pins.
- Mobile Application Display:
 - The Blynk Mobile App is configured with widgets (gauges, graphs, notifications) linked to the same virtual pins.
 - Whenever the ESP8266 updates the data in Blynk Cloud, the mobile app reflects the new values in real time.

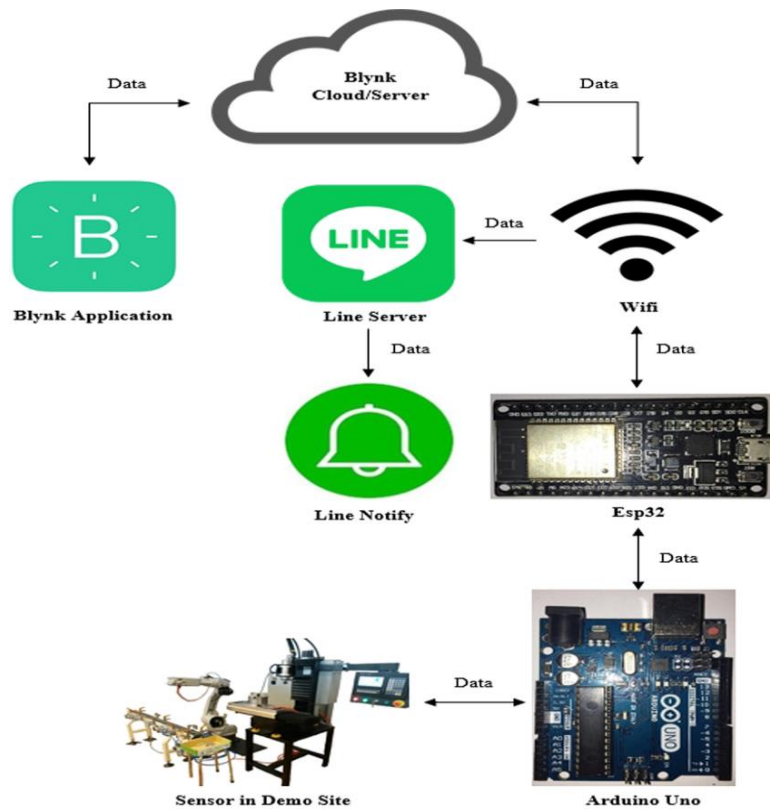


Figure 28: flowchart of Blynk

Blynk Mobile Application Configuration

- Platform: Android / IOS
- Widgets Used:
 - Gauge Widget (V1): Displays the water level as a percentage.
 - Gauge Widget (V2): Displays the turbidity in NTU.
- Update Rate: Real-time updates based on sensor reading frequency.
- Connection: Secure communication with Blynk Cloud over Wi-Fi.

Advantages of Blynk Integration

- Remote Monitoring: View water quality and tank levels from any location.
- User-Friendly Interface: Easy-to-read gauges for clear visualization.
- Scalable: Additional sensors or actuators can be added without major hardware changes.
- Low Power & Efficient: Minimal data usage and power consumption due to optimized IoT protocols

Chapter 6: Design Diagram

PCB Design:

Overview:

To ensure the AquaGuard system is compact, organized, and reliable, a custom PCB was designed using EasyEDA.

The PCB integrates all the critical components needed for the system's operation, reducing the complexity of wiring and improving durability for real-world applications.

Wiring diagram:

The connections between sensors and modules, including the ESP8266 microcontroller, relay, turbidity sensor, pH sensor, and ultrasonic sensor, are shown graphically in the AquaGuard system's wiring diagram, which was made with Fritzing. It offers a clear schematic of the connections between the parts, which facilitates accurate assembly and troubleshooting of the advanced water monitoring system. (Circuits., 2022, June 27)

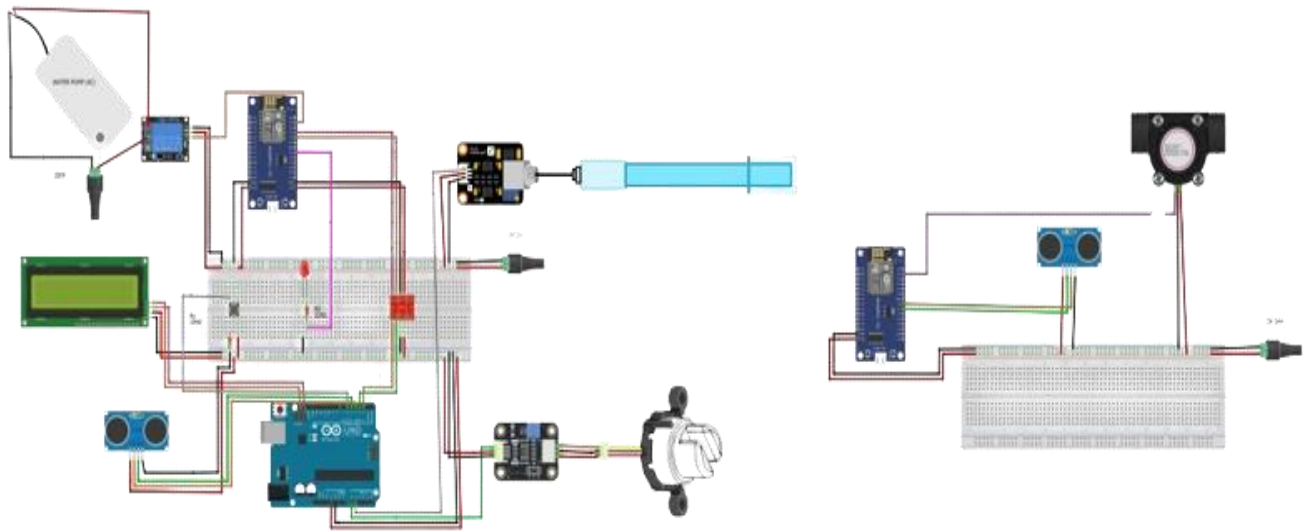


Figure 29:Wiring diagram

This image shows the enclosure containing all the assembled components for the ground tank system

Before/After

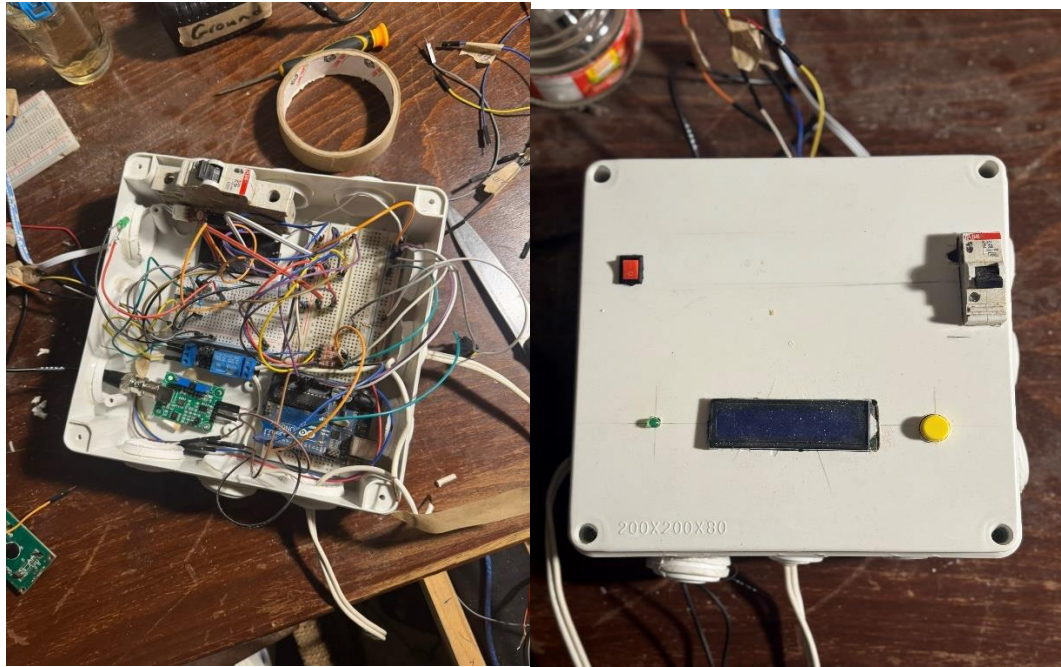
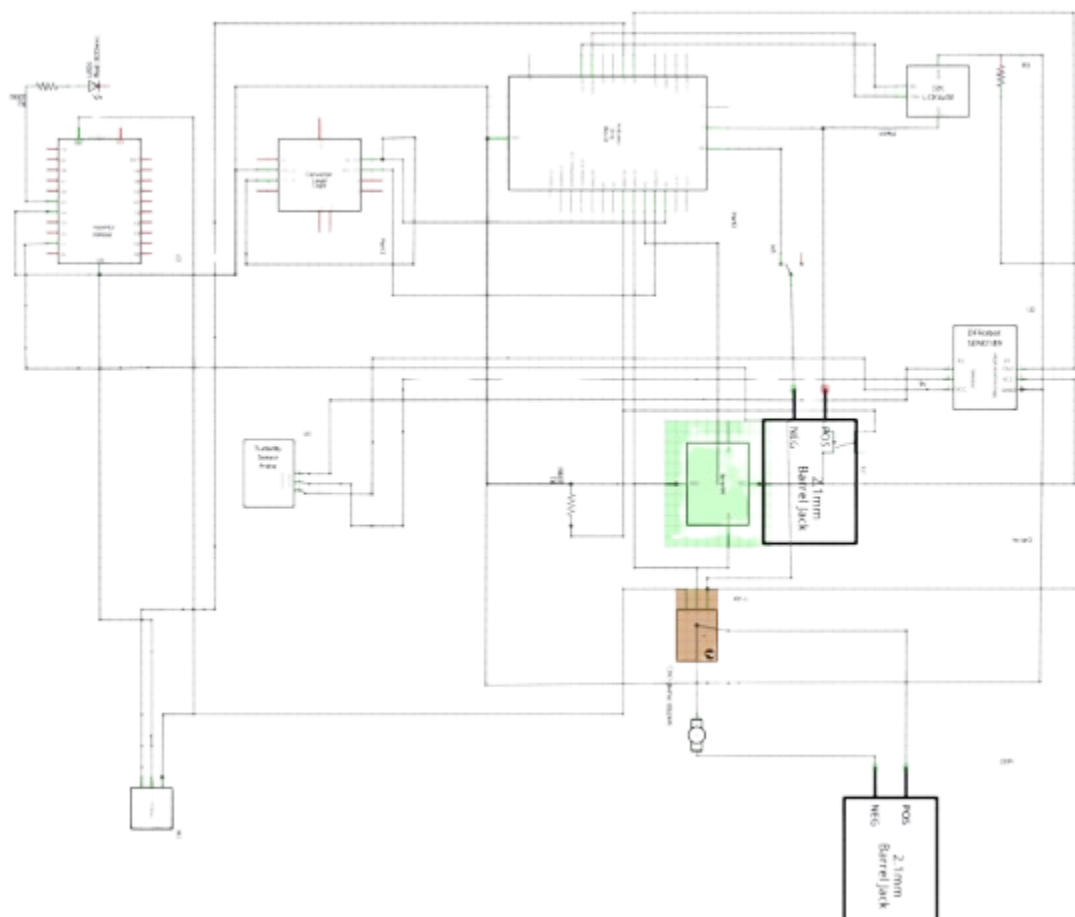


Figure 30 junction box

Schematic: Ground Tank with ESP8266 AND Arduino

*Figure 31: Schematic of ground tank*

Wiring diagram for ground tank:

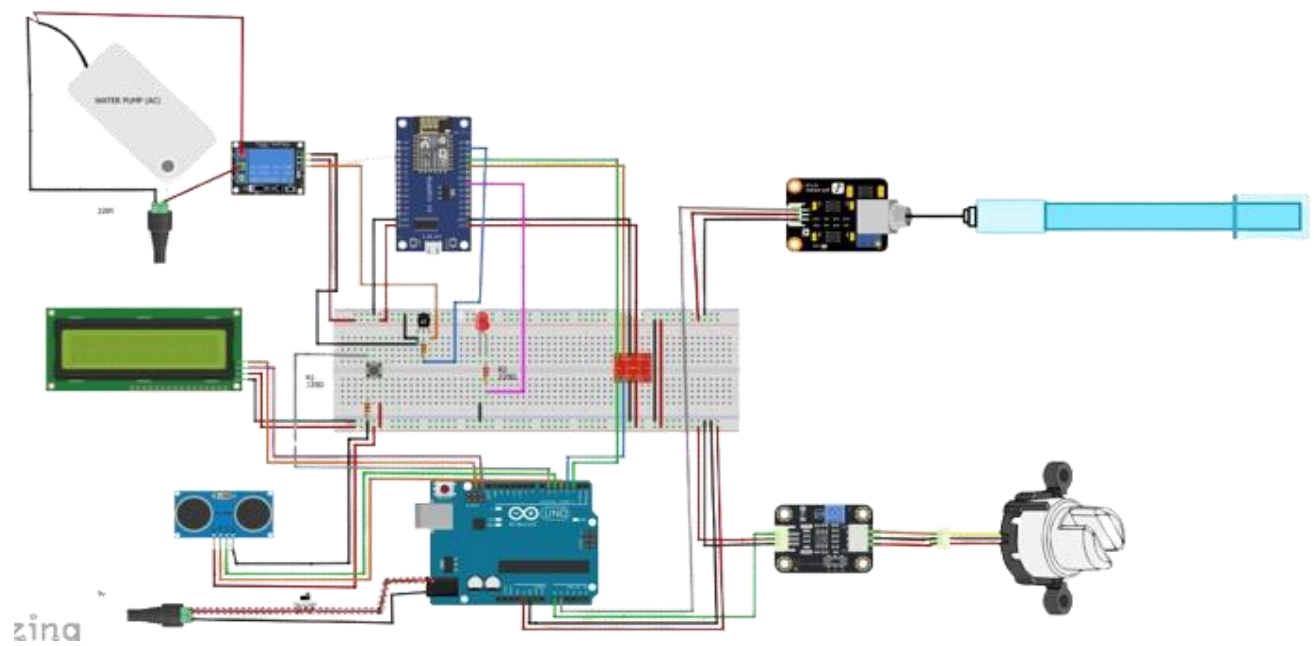


Figure 32: wiring diagram ground tank

Point overview:

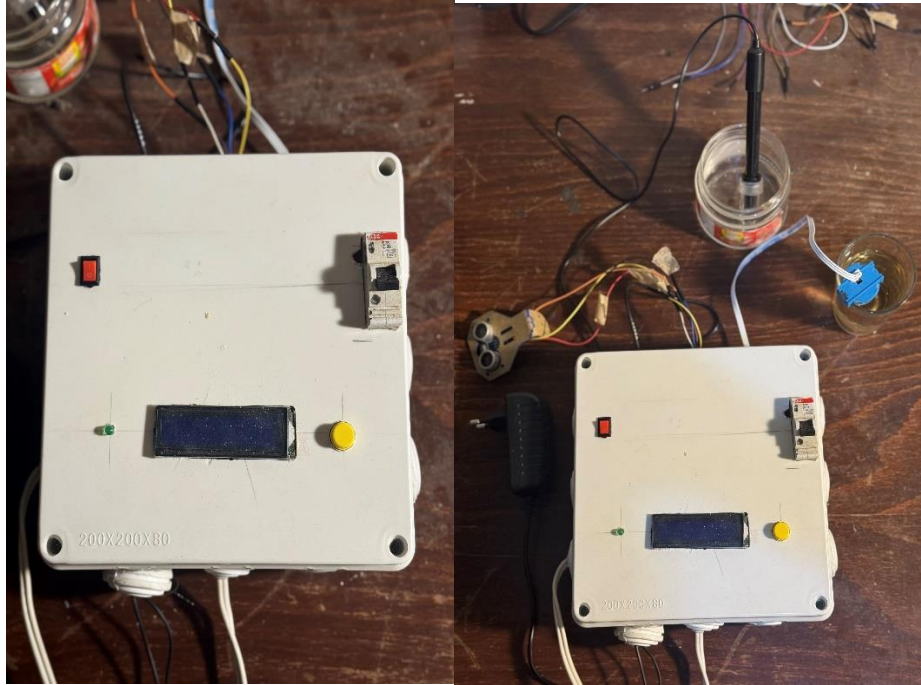


Figure 33 ground box

Schematic & 2D PCB Layout:

The ESP8266 NodeMCU, a single-channel relay module, a logic level converter for 5V-3.3V interfacing, a green status LED, and a pin header for external connections are all included in this 2D schematic of a specially made control board. This EasyEDA-created layout allows clear signal routing, compact integration appropriate for IoT-based automation systems, and safe and effective control of AC or DC loads via Wi-Fi.

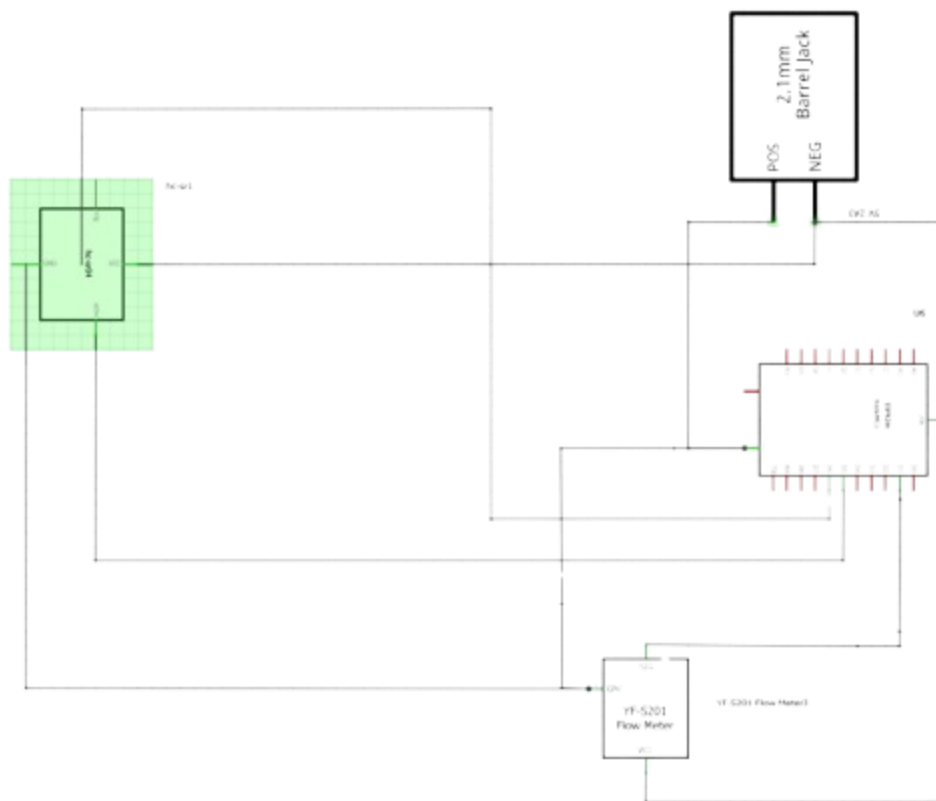


Figure 34: Schematic of roof tank

Wiring Diagram:

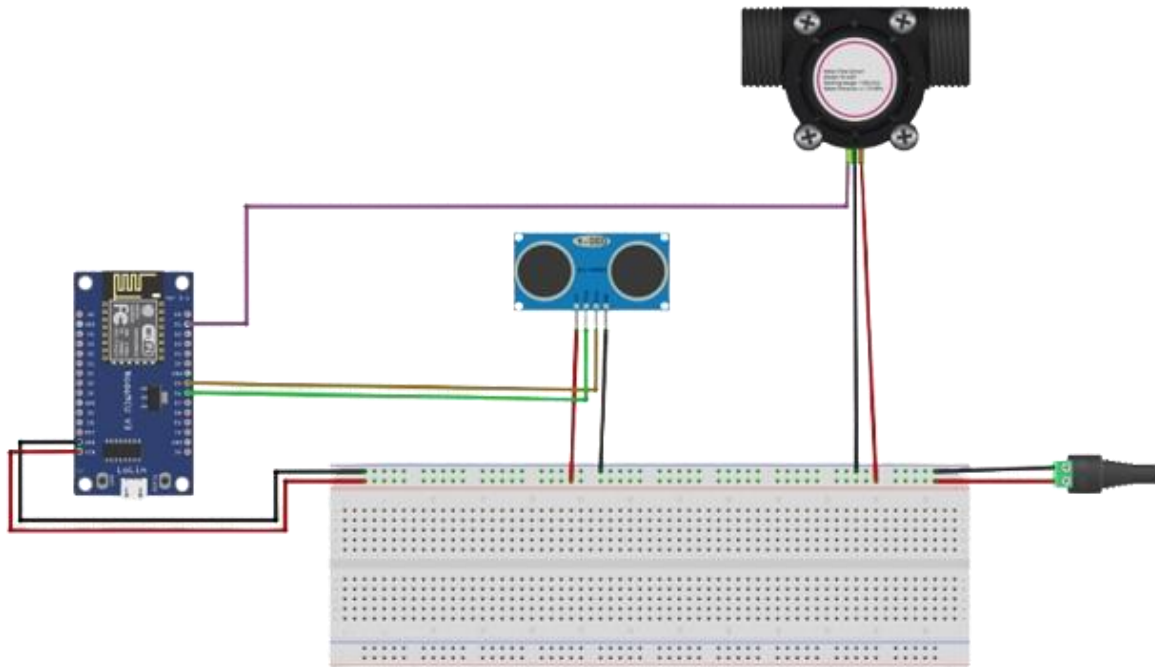


Figure 35:Wiring diagram roof tank

Point overview:

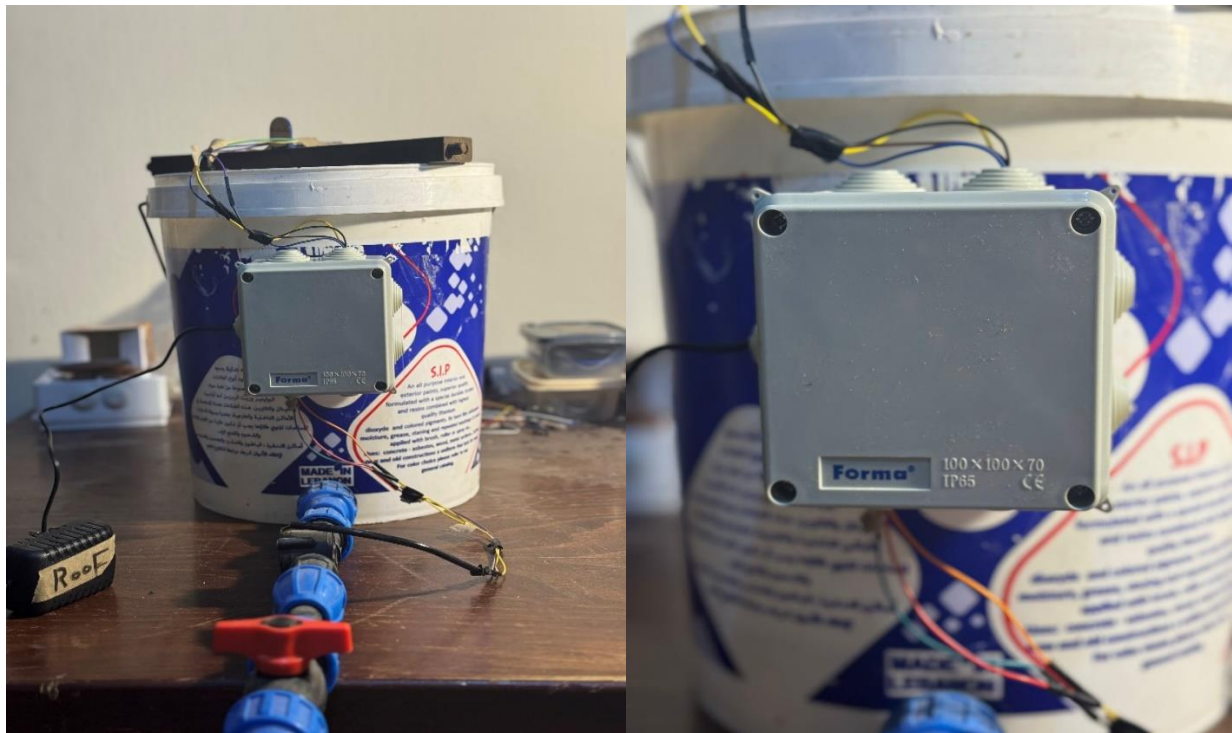


Figure 36 roof box

2D Layout of Controller board

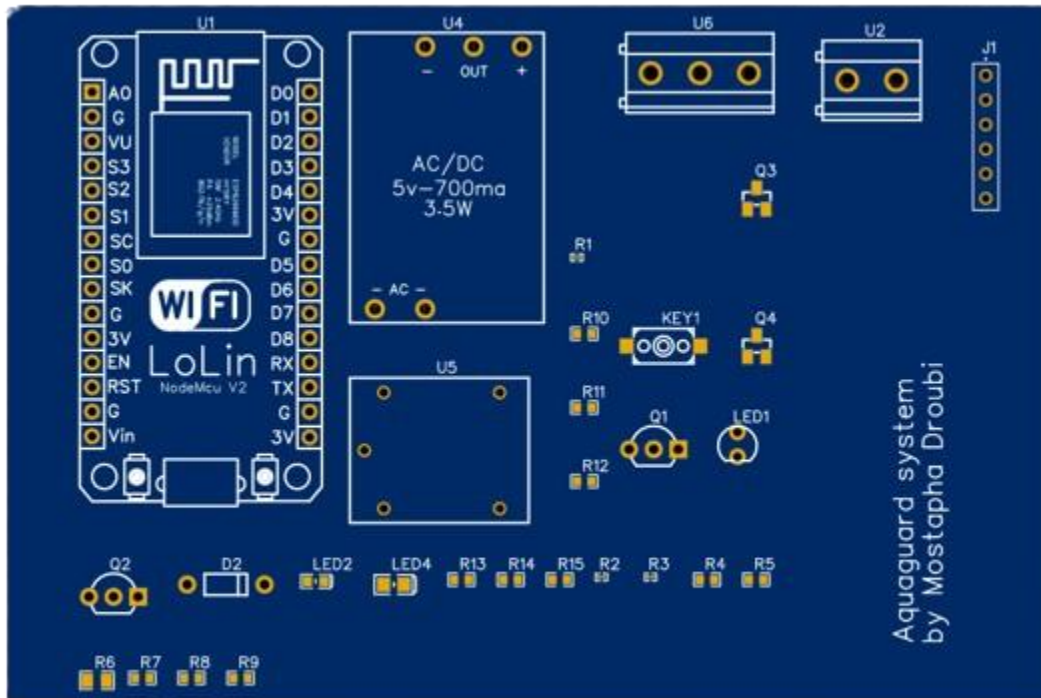


Figure 37: 2D PCB Layout of ESP8266-Based Relay Control Circuit

Schematic of controller board:

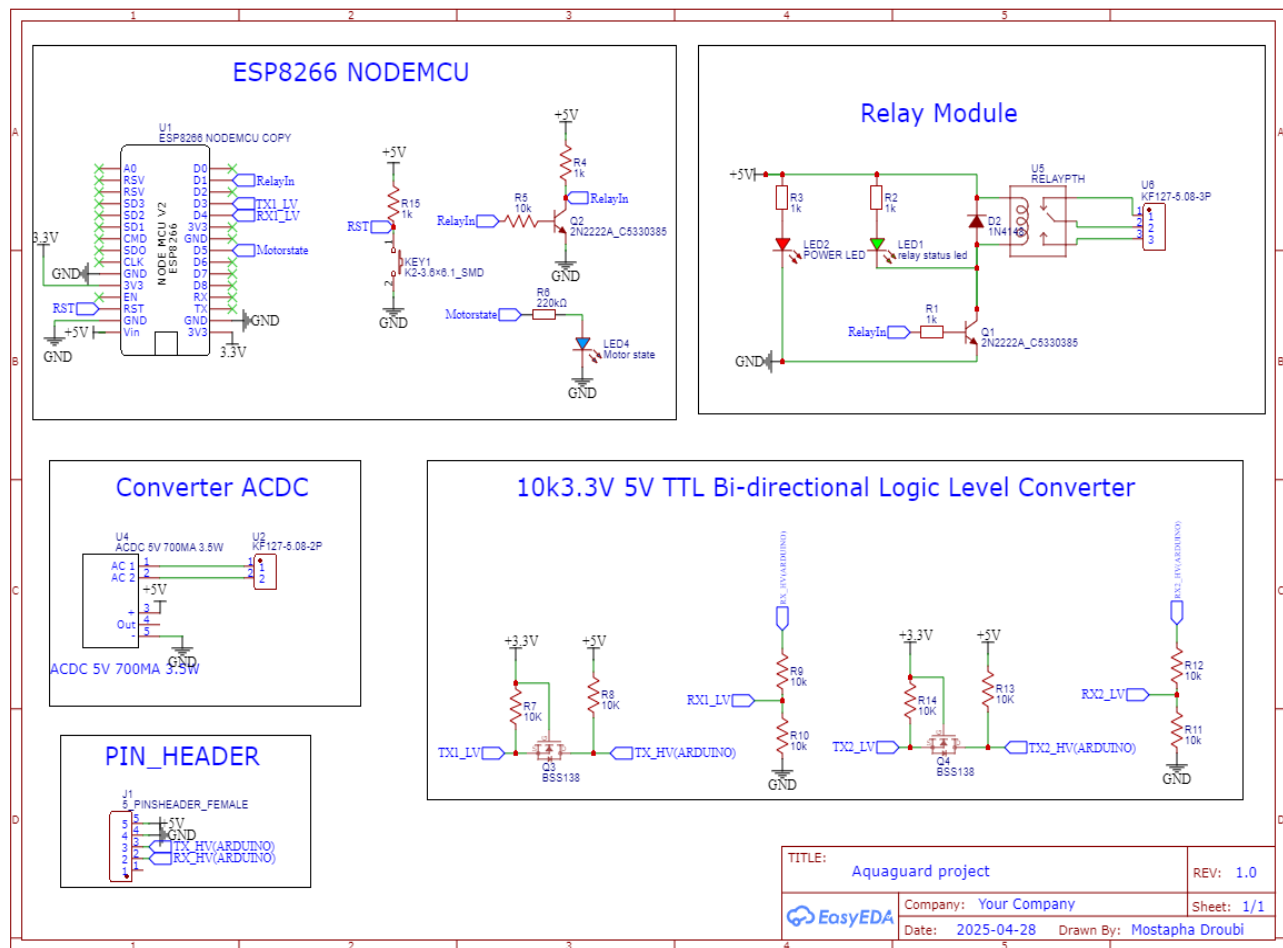


Figure 38 Schematic Diagram of ESP8266-Based Relay Control Circuit

Main Components Integrated on the PCB:

- **ESP8266 NodeMCU Module:** Main controller managing all sensor data and Wi-Fi communication with the Blynk cloud.
- **Relay Driver Circuit:** Controls the motor (water pump) using a 5V relay and a 2N222A NPN transistor. LEDs indicate the relay and motor states.
- **AC to DC Converter:** Provides a regulated 5V DC supply from the AC mains to power the system safely.
- **Bi-directional Logic Level Converter:** Ensures safe communication between the 3.3V ESP8266 and 5V Arduino-based sensors and devices.

- Push Button Reset and Motor Status LEDs: For manual control and system feedback.
- Pin Header Connection: For external Arduino device communication.
- Protection Elements: Flyback diodes, current-limiting resistors, and switching transistors are added for system protection and reliability.

Key Features of the PCB Design:

- Separation between high-voltage (relay, motor) and low-voltage (ESP8266, sensors) sections to ensure safety.
- Compact layout optimized for small enclosures.
- Clear labeling of pins and connectors for easy installation and maintenance.
- Mounting holes for secure attachment inside a waterproof box.
- Designed to reduce human wiring errors and allow future system upgrades easily.

Schematic and PCB Design Tool:

- The schematic and PCB were designed using EasyEDA software.
- The PCB is based on a two-layer design, ensuring a balance between cost and complexity.

Chapter 7: Limitations, appendix, and Future Prospects

Limitations:

Despite its reliability in monitoring water levels, turbidity, and flow, the AquaGuard System has some limitations:

- **Sensor Accuracy:** Ultrasonic and turbidity sensors can be affected by environmental conditions, requiring occasional calibration.
- **Measurement Range:** The sensors are optimized for small to medium setups, limiting industrial-scale applications.
- **Internet Dependency:** System functionality relies on stable Wi-Fi for real-time cloud updates and remote monitoring.
- **System Scope:** Designed mainly for household or small-scale use; larger systems would require further upgrades.

Future Prospects

- **AI-Based Decision Making:** Use of AI to predict water consumption patterns and automate pump operation based on behavior and weather forecasts.
- **Solar Power Integration:** Enabling the system to operate off-grid by incorporating solar panels and battery management.
- **Cloud Data Logging & Analytics:** Adding historical data tracking and visualization through dashboards for long-term water usage analysis.

Appendix

Roof ESP8266 – Water Level & Flow Monitoring:

- Connect to Wi-Fi and Blynk using the authentication token.
- Initialize ultrasonic sensor for water level measurement.
- Initialize flow sensor to measure water flow rate.
- Measure water level using the ultrasonic sensor and convert distance to percentage.
- Send water level data to Blynk Virtual Pin V2.
- If water level is above 80%, send a high-level warning notification to Blynk.
- Measure water flow by counting pulses from the flow sensor and convert to liters per minute.

- Send flow rate data to Blynk Virtual Pin V4.
- If flow rate is greater than 0 but less than 15 L/min, send a low flow alert to Blynk.
- Wait 5 seconds before repeating measurements.

Ground ESP8266 – Relay Control & Water Quality Monitoring

- Connect to Wi-Fi and Blynk using the authentication token.
- Set up serial communication with Arduino to receive sensor data.
- Synchronize roof tank water level from Blynk Virtual Pin V2.
- Read ground tank water level from serial data and send to Blynk Virtual Pin V1.
- Read turbidity values from serial and send to Blynk Virtual Pin V5.
- If turbidity exceeds 1100 NTU, send an alert notification to Blynk.
- Read pH values from serial and send to Blynk Virtual Pin V6.
- If pH is outside safe range (below 6.5 or above 7.9), send a pH alert to Blynk.
- Control relay and motor based on water levels:
- Turn on relay if ground tank level > 10% and roof tank level is between 70% and 90%.
- Turn off relay otherwise.
- Send relay state to Blynk Virtual Pin V3.

Ground Arduino – Water Level, Turbidity & pH Measurement:

- Initialize LCD display and serial communication with ESP8266.
- Measure water level with ultrasonic sensor and convert distance to percentage.
- Read turbidity sensor multiple times, filter readings by averaging, and convert to NTU.
- Read pH sensor multiple times, average readings, convert voltage to pH value, and determine water pH state (acidic, neutral, or alkaline).
- Detect button press with debounce logic to cycle LCD display through:
 - Welcome message
 - Water level
 - Turbidity
 - pH value and state
- Update LCD display to show current mode data.
- Send water level, turbidity, and pH readings to Ground ESP8266 via serial with the formats:
 - "WATER:" + water level percentage
 - "TURB:" + turbidity NTU
 - "PH:" + pH value

Cost of elements

Element	Number of pieces	Price for pieces \$	Total
Arduino uno r3	1	15	15
Esp8266	2	12	12
LCD I2C	1	7	7
Ultrasonic sensor	2	4	4
Turbidity sensor	1	15	15
Logic level shifter	1	2.5	2.5
1 channel Relay module	1	1.5	1.5
Push button	1	0.2	0.2
Ph sensor	1	25	25
Flow sensor	1	8	8
Water pump Pedrollo	1	30	30
Wire	60	0.1	6
Resistor(210ohm)	2	0.04	0.08
Adapter 5V 2A	1	2	2
Adapter 9V 2A	1	2	2
Total			130.28\$

Table 9

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